INTERNSHIP FINAL PROJECT SHOWCASE

Car Showroom Sales Analysis & Price Prediction

Data-driven inventory intelligence, customer behavioral insights, and supervised machine learning valuation models.

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Designation
Data Analytics Intern

Domain
Automotive Retail Analytics

CONTEXT & ARCHITECTURE

Project Scope & Core Problem Statement

Sales Performance

Uncovering high-margin brand and category contributors.

Customer Behavior

Fuel preference and transmission adoption patterns.

Valuation Engine

Machine learning predictive pricing pipeline.

EXECUTIVE METRICS

Operational Footprint & Cumulative Yield

600

Total Units Analyzed

Zero missing entries

₹1.007B

Gross Realized Turnover

₹1.67M average ticket size

₹129.8M

Net Operating Profit

12.89% average dealer spread

Unit Economics Benchmark

Avg Purchase: ₹1,462,425 → Avg Selling: ₹1,678,877

MARKET DEMAND

Volume Leaders & Top Models

Top Brands by Volume

Toyota	110 units
Honda	106 units
Maruti Suzuki	103 units

Top Preferred Models

Toyota Innova	45 units
Hyundai Creta	42 units
Honda Amaze	40 units

MARGIN CONTRIBUTION

Category Profitability Distribution

45.3% Margin Share

SUV Segment

₹58,865,158

Highest gross profit contributor.

44.7% Margin Share

Hatchback Segment

₹58,028,048

High volume turnover engine.

10.0% Margin Share

Sedan Segment

₹12,978,302

Niche premium customer base.

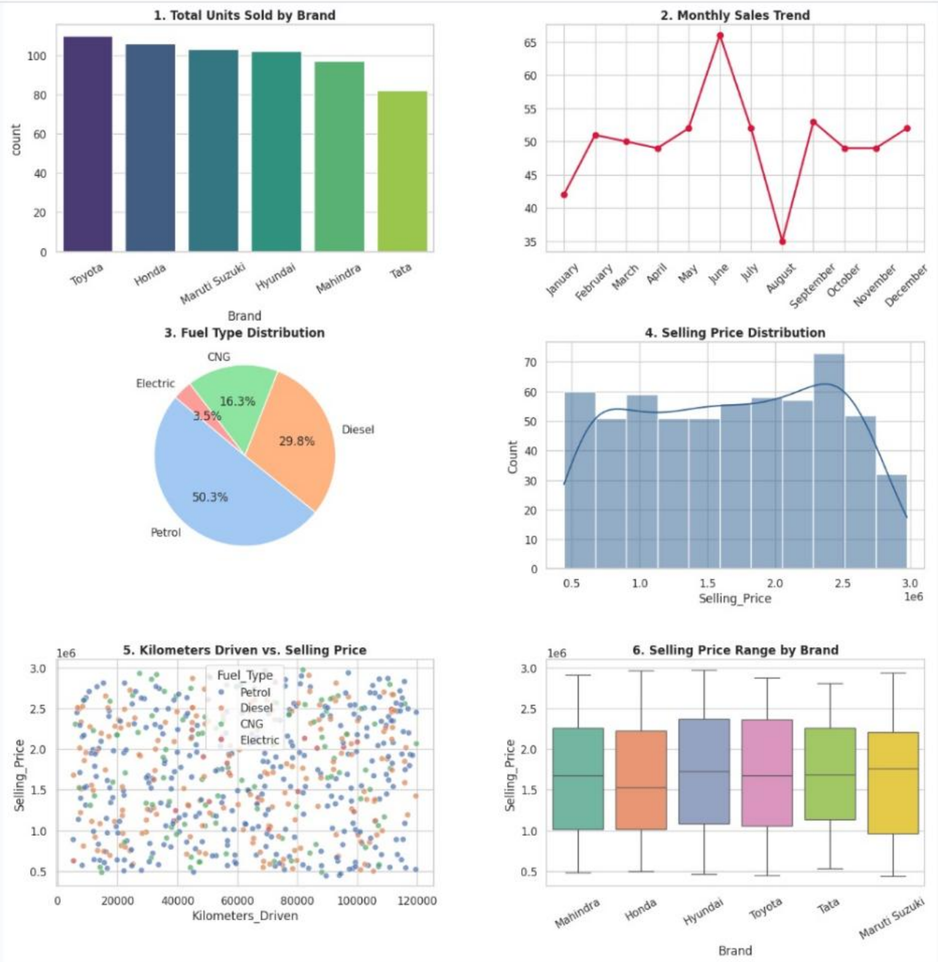
TEAM PERFORMANCE

Sales Representative Throughput

Representative	Revenue Generated	Contribution	Primary Driver
Sneha Patil	₹266,930,545	26.5%	Premium SUV Volume
Rahul Sharma	₹260,771,833	25.9%	Balanced Portfolio
Priya Kulkarni	₹249,457,331	24.8%	Family Sedan Conversions
Amit Deshmukh	₹230,166,863	22.8%	Hatchback Velocity

EXPLORATORY GRAPHS

Visual Trends & Behavioral Insights



June Annual Spike

Peak sales reached 66 units.

Monsoon Contraction

August demand falls to 35 units.

Petrol Dominance

Petrol holds 50.3% market share.

DATA PIPELINE

Feature Engineering & Data Preparation

STEP 01

Vehicle Age

Selling year minus manufacturing year.

STEP 02

Profit Margin

Selling price minus purchase price.

STEP 03

Temporal Extraction

Month and seasonal cycle isolation.

STEP 04

One-Hot Encoding

Categorical encoding avoiding dummy traps.

EVALUATION & ACCURACY

Model Evaluation & Benchmarking

Model Architecture	MAE (₹)	RMSE (₹)	R^2 Score	Status
Linear Regression	₹48,212.26	₹62,059.06	0.9923	**Champion Model**
Random Forest Regressor	₹51,079.18	₹67,229.17	0.9909	Runner-up
Decision Tree Regressor	₹63,825.79	₹85,305.36	0.9854	Baseline Model

Linear Regression delivered the lowest variance without overfitting risk.

PREDICTION PRECISION

Winning Model Accuracy

99.23%

Variance explained by the Linear Regression model

Mean Absolute Error: ₹48,212 on ₹1.67M vehicles

ACTION PLAN

Operational Recommendations

Inventory Sourcing

Prioritize high-demand Toyota and Hyundai units.

Monsoon Campaigns

Incentivize August sales to counter season drop.

Automated Pricing

Deploy ML valuation engine into showroom CRM.

PROJECT COMPLETE & VERIFIED

Thank You

Questions & Guide Feedback

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